

Aditya Puntambekar

Generative AI & NLP Engineer | RAG Systems | Neural Networks

Pune, Maharashtra, India | aditya23703@gmail.com | +91 7208275414 | github.com/Adityapuntambekar | linkedin.com/in/aditya-puntambekar | BSc Data Science Student

Profile

BSc Data Science student and Generative AI engineer specializing in Retrieval-Augmented Generation, LLM application pipelines, and NLP systems, with applied research experience in spiking neural networks for real-time perception. Built and shipped portfolio-grade systems spanning agentic RAG, conversational AI, and neuromorphic point-cloud processing.

Core Skills

Generative AI & LLMs	LangGraph, LangChain, Prompt Engineering, Corrective/Agentic RAG, Google Gemini API, OpenAI API, Prompt-Injection Defense
RAG & NLP	FAISS, BM25, Hybrid Retrieval, Cross-Encoder Reranking, BGE Embeddings, Semantic Caching, RAGAS Evaluation, Document Parsing & Chunking
Neural Networks & SNN	Spiking Neural Networks, Neuromorphic Computing, Point-Cloud/LiDAR Processing, PyTorch, Sensor Fusion (Kalman Filters)
ML / Deep Learning	scikit-learn, PyTorch, Federated Learning (Flower), Homomorphic Encryption (TenSEAL/CKKS), Imbalanced-Data Evaluation
Languages & Tools	Python, JavaScript, C++, FastAPI, Streamlit, Gradio, Docker, Langfuse, Git/GitHub, Google Colab

Featured Projects

FinRAG — Agentic RAG System for Financial Filings

Python, LangGraph, Google Gemini, FAISS, BM25, Streamlit

- Built a retrieval-augmented generation pipeline over Apple and Microsoft SEC 10-K filings using hybrid dense (BGE embeddings + FAISS) and sparse (BM25) retrieval with cross-encoder reranking for higher answer precision.
- Designed a LangGraph-based corrective RAG loop that self-evaluates retrieved context and re-retrieves on low-confidence answers, reducing hallucination in financial Q&A.
- Implemented prompt-injection safeguards, semantic response caching, and Langfuse tracing for full pipeline observability, plus a RAGAS-based evaluation harness benchmarking answer faithfulness and relevance.
- Shipped an interactive Streamlit demo for end-to-end document ingestion, query, and citation-grounded response generation.

[View on GitHub → github.com/Adityapuntambekar/FinRAG](https://github.com/Adityapuntambekar/FinRAG)

AVRLM — Spiking Neural Network for Real-Time LiDAR Perception

Python, PyTorch, Spiking Neural Networks, Semantic KITTI, Streamlit

- Developing a spiking neural network (SNN) point-model for radar-LiDAR sensor fusion, achieving 261 FPS on point-cloud inference for real-time autonomous perception.
- Integrated the Semantic KITTI dataset (21 sequences with calibration and timestamp data) to validate the model on real-world driving scenes up to 100 m range, targeting reliability standards for defense-oriented (DRDO) evaluation.
- Built synthetic LiDAR/radar data generation and a KITTI data-loading pipeline to support semantic training and evaluation alongside real sensor data.
- Designing a real-time GUI dashboard to visualize spiking-model inference and 3D point-cloud reconstruction; authoring an accompanying research paper.

[View on GitHub → github.com/Adityapuntambekar/avrlm_dekstop_app](https://github.com/Adityapuntambekar/avrlm_dekstop_app)

HELIX — V2X Emergency-Vehicle Path-Clearing Simulator

Python, Streamlit, Plotly, SAE J2735

- Prototyped the messaging layer of a V2X emergency-vehicle system, implementing SAE J2735-compliant BSM/SRM message construction and geofenced signal preemption logic ahead of embedded (ESP32) porting.

- Built a 5-state Extended Kalman Filter sensor-fusion layer, a synthetic route generator, and a mock Roadside Unit (RSU) to simulate real-time vehicle-to-infrastructure communication.
- Delivered a Streamlit + Plotly dashboard for live geofence and trajectory visualization, backed by 119 passing unit and integration tests.

[View on GitHub → `github.com/Adityapuntambekar/helix-v2x-simulator`](https://github.com/Adityapuntambekar/helix-v2x-simulator)

FedRare — Privacy-Preserving Federated Learning for Rare-Disease Diagnosis

Python, PyTorch, Flower, TenSEAL (CKKS), Docker

- Built a federated learning system training a diagnostic MLP across four simulated hospital nodes on the UCI Cirrhosis Patient Survival dataset without centralizing patient data.
- Designed a custom HomomorphicFedAvg strategy that aggregates CKKS-encrypted model updates server-side using TenSEAL homomorphic encryption, preserving end-to-end data privacy.
- Benchmarked encrypted vs. plaintext aggregation, achieving comparable accuracy (~89-92%) while quantifying the privacy-performance trade-off in a Dockerized multi-hospital simulation.

[View on GitHub → `github.com/Adityapuntambekar/fedrare-privacy-preserving-fl`](https://github.com/Adityapuntambekar/fedrare-privacy-preserving-fl)

Education & Achievements

- BSc Data Science and Big Data Analytics (Hons) — 2024 - 2027 (Currently Pursuing).
- Finalist, AWS Hackathon; participant, HackMIT, IIT Kharagpur Data Science Hackathon, and Smart India Hackathon (SIH).